tf.compat.v1.mixed_precision.LossScale Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/mixed_precision/LossScale

> *Base class for all TF1 loss scales.*

Function Signatures

```
tf.compat.v1.mixed_precision.LossScale()
```

Code Examples

```
tf.compat.v1.mixed_precision.LossScale()
```

```
tf.compat.v1.mixed_precision.LossScale()
```

```
loss = ...
loss *= loss_scale
grads = gradients(loss, vars)
grads /= loss_scale
```

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loss = ...
loss *= loss_scale
grads = gradients(loss, vars)
grads /= loss_scale
```

```
@classmethod
from_config(
config
)
```

```
@classmethod
```

```
from_config(
config
)
```

```
@abc.abstractmethod
get_config()
```

Documentation

Base class for all TF1 loss scales.

View aliases

Compat aliases for migration

See
Migration guide for
more details.

tf.compat.v1.mixed_precision.experimental.LossScale,
tf.compat.v1.train.experimental.LossScale

This is an abstract base class, so you cannot instantiate it directly.
Instead, use one of its concrete subclasses:

- tf.compat.v1.mixed_precision.DynamicLossScale
- tf.compat.v1.mixed_precision.FixedLossScale

Loss scaling is a process that multiplies the loss by a multiplier called the loss scale, and divides each gradient by the same multiplier. The pseudocode for this process is:

Mathematically, loss scaling has no effect, but can help avoid numerical underflow in intermediate gradients when float16 tensors are used for mixed precision training. By multiplying the loss, each intermediate gradient will have the same multiplier applied.

Instances of this class represent a loss scale. Calling instances of this class returns the loss scale as a scalar float32 tensor, while method update() updates the loss scale depending on the values of the gradients. Optimizers use instances of this class to scale loss and gradients.

In most functions that accept a LossScale, you can also pass an int (such as

8) to create a FixedLossScale or the string "dynamic" to create a dynamic loss scale.

Methods

from_config

View source

Creates the LossScale from its config.

get_config

View source

Returns the config of this loss scale.

update

View source

Updates the value of the loss scale.

The loss scale will be potentially updated, based on the value of grads.

The tensor returned by calling this class is only updated when this function is evaluated.

In eager mode, this directly updates the loss scale, so that calling __call__ will return the newly updated loss scale. In graph mode, this returns an op that, when evaluated, updates the loss scale.

This function also returns a should_apply_gradients bool. If False, gradients should not be applied to the variables that step, as nonfinite gradients were found, and the loss scale has been be updated to reduce the chance of finding nonfinite gradients in the next step. Some loss scale classes will always return True, as they cannot adjust themselves in response to nonfinite gradients.

When a DistributionStrategy is used, this function may only be called in a cross-replica context.

__call__

View source

Returns the current loss scale as a scalar float32 tensor.